

Paper #9 v11 outline — The Arithmetic Triangle of Curved Space (K53 cascade extension to k=24)

2026-05-18

Paper #9 v11 — The Arithmetic Triangle of Curved Space (with K53 cascade extension)

v0.1 outline scope

This outline captures the v11 additions to the existing v10 paper structure. Substantive content for new sections to be filled in by Elie (Three Theorems lane) post-K52a. Lyra/Grace pre-staged operator-level framework available for cross-references.

New content in v11 (relative to v10)

Section 9.5 NEW — Cascade extension k=21..24 (K53 D-tier structural law)

Content to add: - Elie Toy 3051 extraction methodology (Toy 1610 re-run, n=44 dps=3200 upgrading prior dps=1600) - Verification table: $a_{21}, a_{22}, a_{23}, a_{24}$ explicit values + ratio matches against Toy 2994 closed-form prediction $a_k/a_{k-1} = -k(k-1)/(2 \cdot n_C) - 11/11$ ratio matches (Toy 3051 SCORE) - Known speaking-pair levels k=5,6,10,11,15,16,20,21 confirmed; k=22 (Toy 1610 zero-margin upgrade to +5 margin); k=23 and k=24 NEW levels (k=24 at data limit margin +1) - 24 consecutive levels of mechanism-confirmed BST primary heat kernel structure at level (3) integrated SD on D_{IV}^5

Section 9.6 NEW — K53 audit verdict + scope qualifier (CRITICAL discipline)

Content to add: - K53 PROMOTED to D-tier structural law per Keeper governance ruling 2026-05-18 (audit doc: notes/K53_Three_Theorems_Cascade_Extension.md) - Promotion criteria met (Cal Criterion 1 + 2 + 3): - Pre-staged formula $\times$ forward verification (Toy 2994 closed form pre-extraction) - Sample size 24 levels (4 full speaking-pair periods + extension) - Margin discipline (k=22 zero-margin $\rightarrow$ +5 margin upgrade) - Independent extraction methodology (dps=1600 $\rightarrow$ dps=3200) - Scope qualifier LOAD-BEARING (per Keeper governance): closed form $a_k/a_{k-1} = -k(k-1)/(2 \cdot n_C)$ applies SPECIFICALLY to: - Three Theorems cascade - Level (3) integrated Seeley-DeWitt (NOT point-trace levels (1)/(2)) - On D_{IV}^5 (NOT other geometries) - Through k=24 (extension beyond k=24 is open question) - Universality beyond level (3), beyond D_{IV}^5 , or beyond k=24 — all open questions, do NOT claim - External-survivability framing required: present forward-locked methodology PROMINENTLY (pre-staged formula $\rightarrow$ forward verification), NOT “we observed a pattern”

Section 9.7 NEW — Connection to point-trace operator framework (Keeper three-level discipline)

Content to add: - Keeper’s three-level operator-trace framework (2026-05-18 PM clarification): - Level (1): $\text{Tr}(D^{2^k})$ at one point of D_{IV}^5 , Lie-algebraic (Lyra T2376: $32 \cdot 10^k$) - Level (2): Heat-kernel-coefficient $\text{Coeff}_n = \text{Tr}(D^{2^n})/n!$ at origin (Grace T2375: $32 \cdot 10^n/n!$; Lichnerowicz scalar: $32 \cdot (75/4)^n/n!$) - Level (3): Integrated Seeley-DeWitt a_n (Elie SP-3 Three Theorems cascade) - Lichnerowicz cleanly bridges (1) $\leftrightarrow$ (2) at point-trace level (Lyra T2378 + Toy 3050, Monday afternoon pre-execution) - Bridge (1)/(2) $\rightarrow$ (3) requires multi-week Bergman volume integration + $\hat{A}$ -genus corrections - This v11 paper documents the level (3) cascade closure;

level (1)/(2) operator framework is companion content in Paper #118 v0.2 Section 9 (Bergman Dirac) + Paper #115 v0.5 (architectural framework)

Section 9.8 NEW — Methodology contribution: pre-registration + forward verification

Content to add: - Casey's hunting principle: "deviations locate boundaries" — cascade survives → no boundary at $k \leq 24$ - Pre-registration discipline: closed form Toy 2994 → pre-staged Toy 1610 + 3051 → forward verification - Audit-chain context: K53 PROMOTED is one of 8+ calibrations in 36 hours via team self-audit discipline; "audit chain produces honest tier labels by construction" - Cal's coincidence-filter discipline (6 named failure modes + Rule 6 corollary + Strict Context-Counting Protocol): K53 passes Test B (pre-staged $\times$ forward verification $\times$ sample size $\times$ margin discipline) cleanly

Appendix C-extension NEW — Future cascade extension to $k \geq 25$

Content to add: - Cascade extension to $k=25+$ requires more independent n-values ($k=24$ at data limit margin +1) - SP-3 program continues toward $k \geq 25$ at sustainable compute pace - Per Keeper post-K53: if cascade continues to $k=25-30$, more compute resources warrant scaling - Per Casey's "deviations locate boundaries" — boundary at some $k_{\max}$ would itself be a major structural finding

Existing v10 content to retain (per K53 ruling, no rewrite of v10 content)

- §1-§8 unchanged (introductory, Three Theorems statement, low-k verification through $k=20$)
- §9.1-§9.4 retained (Speaking-pair period $n_C=5$, four full periods through $k=20$, cyclotomic tameness)
- §9.2a gauge hierarchy headline retained
- ($C=12$, $D=0$) factorization retained

Strategic role

Paper #9 v11 is the first of the 2026-spring papers to land a K-audited D-tier structural-law claim (K53). Companion papers in the spring portfolio (per Keeper's 4-paper coherent set): - Paper #9 v11: heat kernel cascade level (3) structural-law (Elie lead, this paper) - Paper #115 v0.5: BST architectural framework (Grace lead, complete with K50 L1 mediated tier) - Paper #118 v0.2: Bergman Dirac operator at level (1)/(2) (Lyra lead, ready for Cal grade-pass) - Paper #119: SP-29 dual-falsifier program (three-CI v0.3 paper-grade) - Paper #120 v0.2: G and inertia substrate-mediated (three-CI v0.2 merged outline)

Honest scoping for v0.2+

Closed by v0.1 outline (this filing): - New section structure for v11 (Sections 9.5-9.8 + Appendix C-extension) - K53 D-tier ruling captured with explicit scope qualifier - Point-trace operator framework cross-reference (Paper #118 v0.2 + #115 v0.5) - Methodology discipline framing (audit-chain + coincidence-filter)

Open for v0.2 substantive drafting (Elie lane post-K52a): - Section 9.5 verification table (Toy 3051 explicit data + ratio matches) - Section 9.6 K53 audit discipline narrative - Section 9.7 three-level framework explication - Section 9.8 methodology contribution paragraphs - Appendix C-extension future-work scope

v0.3 plan: Cal gate-pass + Grace catalog integration + cross-reference audit.

v1.0 submission target: J. Spectral Theory (per v10) OR Inventiones (per cascade significance). Editor's call.

— Lyra, v0.1 outline filed per Casey "do all" directive 2026-05-18 PM. Substantive content drafting is Elie's Three Theorems lane post-K52a.